Literature Review & System Design

Voice-Based Speaker Authentication System

Final Year CSE Project

1. Project Overview

This project implements a Voice-Based Speaker Authentication System capable of two core tasks:

- Speaker Identification (1-to-N): Given an unknown voice recording, determine *who* is speaking out of a set of registered speakers.
- Speaker Verification (1-to-1): Given a voice recording and a *claimed* identity, determine whether the claim is genuine or fraudulent.

The system uses a pretrained state-of-the-art neural network (ECAPA-TDNN) to extract biometric "voiceprints" from audio, and compares them using cosine similarity for both tasks.

2. Literature Review

The following research papers were studied to understand the problem domain and justify the design decisions made in this project.

Paper 1: Gaussian Mixture Models for Speaker Verification

Citation: Reynolds, D. A., & Rose, R. C. (1995). Robust text-independent speaker identification using Gaussian mixture speaker models. IEEE Transactions on Speech and Audio Processing, 3(1), 72–83.

What this paper says:
This is one of the foundational papers in speaker recognition. It proposed using Gaussian Mixture Models (GMMs) to statistically model the acoustic properties of each person's voice. A GMM is essentially a mathematical "recipe" that describes what a speaker's voice typically looks like when broken into small pieces. To identify who is speaking, you compare a new recording against every speaker's GMM model and see which one best "fits" it — like comparing a fingerprint to a database.

Key Limitation: The system required a lot of adaptation data for each speaker and was fragile in noisy environments or when the audio quality changed (e.g., different microphone, room echo). It also took a long time to run the comparison against every model.

What we learned from it: This paper established the baseline approach to speaker recognition that was used for two decades. Understanding it helped us appreciate why the deep learning approach used in our project is so

much better — it learns a compact fingerprint (embedding) that is inherently more noise-robust.

Paper 2: i-Vector Framework for Speaker Recognition

Citation: Dehak, N., Kenny, P. J., Dehak, R., Dumouchel, P., & Ouellet, P. (2011). Front-end factor analysis for speaker verification. IEEE Transactions on Audio, Speech, and Language Processing, 19(4), 788–798.

What this paper says:
The i-vector framework was a massive leap forward. Instead of storing a whole statistical model per speaker (like GMM), it proposed a technique to compress all the acoustic information about a speaker into a single, compact numeric vector (the "i-vector") of fixed size (usually 400–600 numbers). The key insight was to find a mathematical "total variability space" that separates what makes your voice unique from what is just noise from the environment or the recording device.

Layman Explanation: Imagine every human voice can be described by a point in a giant 3D map of voices. Your voice might be at coordinate (3.2, -1.8, 5.5) and Omar's might be at (8.1, 2.4, -3.0). The i-vector is that coordinate point. To identify you, the system just checks which stored coordinate is closest to the new one.

What we learned from it: The i-vector concept is the direct predecessor of the "embedding" concept we use today. Our project uses a deep learning model to produce a 192-dimensional embedding vector — it's the same fundamental idea, but done far more powerfully with a neural network. This paper helped us understand the mathematical justification for why comparing vectors (and specifically using cosine similarity) works for speaker matching.

Paper 3: X-Vectors — Deep Learning for Speaker Recognition

Citation: Snyder, D., Garcia-Romero, D., Sell, G., Povey, D., & Khudanpur, S. (2018). X-vectors: Robust DNN embeddings for speaker recognition. Proceedings of the 2018 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), pp. 5329–5333.

What this paper says:
This paper introduced "x-vectors" — a deep learning alternative to i-vectors. A Time-Delay Neural Network (TDNN) is trained on thousands of hours of speech from thousands of speakers. Instead of handcrafting the mathematical model, the neural network learns what makes voices unique from data. The network includes a clever "statistical pooling" layer that aggregates information across the entire utterance (not just a single frame), producing one fixed-size vector for any length of audio clip.

A critical contribution was proving that data augmentation (artificially adding noise, reverb, and music to training recordings) dramatically improved robustness. The x-vector system significantly outperformed the i-vector system on real-world tasks.

What we learned from it: X-vectors are the direct architectural ancestor of ECAPA-TDNN (which we use). Understanding x-vectors explained why we chose a TDNN-based model. The statistical pooling concept is what allows our system to handle audio clips of different lengths (our recordings range from 2.4s to 5.3s) and still produce a fixed 192-dimensional embedding. This paper confirmed our choice of architecture.

Paper 4: ECAPA-TDNN — Our Core Model

Citation: Desplanques, B., Thienpondt, J., & Demuynck, K. (2020). ECAPA-TDNN: Emphasized Channel Attention, Propagation and Aggregation in TDNN Based Speaker Verification. Proc. Interspeech 2020, 3830–3834. DOI: 10.21437/Interspeech.2020-2650. arXiv:2005.07143

What this paper says:
This is the model we directly used in our project. ECAPA-TDNN improves upon x-vectors through three key innovations:

1. **Channel Attention (E):** The network learns to focus on the most speaker-discriminative frequency channels. Think of it like the network automatically knowing that low-frequency vowel sounds are more useful for identifying a person than high-frequency consonants.
2. **Propagation:** Features are passed across multiple layers to allow the network to capture both short-term patterns (individual sounds) and long-term patterns (speaking rhythm, intonation style).
3. **Aggregation (A):** An Attentive Statistics Pooling layer is used — instead of treating every frame equally when creating the final embedding, it assigns higher importance to the most informative moments in the speech.

The model was trained on the massive VoxCeleb dataset — over 7,000 real-world speaker identities with audio from interviews, YouTube videos, and podcasts. This is why it generalizes so well to our own voices that it has never been trained on.

What we learned from it: This paper directly justified our entire implementation. We used the pre-trained ECAPA-TDNN model from SpeechBrain (trained on VoxCeleb) and applied it in a "zero-shot" style — meaning we never retrained the model on our own data. We simply asked it to extract embeddings for our speakers' voices and averaged those embeddings to create biometric profiles. The 100% accuracy result on our 27 test recordings validated that this approach works even for a small, custom dataset.

Paper 5: SpeechBrain — The Toolkit We Used

Citation: Ravanelli, M., Parcollet, T., Plantinga, P., Rouhe, A., Cornell, S., Lugosch, L., ... & Bengio, Y. (2021). SpeechBrain: A General-Purpose Speech Toolkit. arXiv:2106.04624.

What this paper says:
SpeechBrain is an open-source speech processing toolkit built on PyTorch. It was designed so that researchers and engineers could use state-of-the-art speech models without having to re-implement complex architectures from scratch. It provides pretrained models for speaker recognition, speech recognition, speech enhancement, diarization, and more — all through a simple, unified Python API.

What we learned from it / why we used it: Using SpeechBrain dramatically reduced the complexity of our implementation. Instead of building and training ECAPA-TDNN ourselves (which would require thousands of hours of data and weeks of GPU compute time), we called EncoderClassifier.from_hparams("speechbrain/spkrec-ecapa-voxceleb") to download and use the pretrained model in 2 lines of code. This is standard professional practice in modern ML engineering — using proven pretrained models as a foundation (Transfer Learning).

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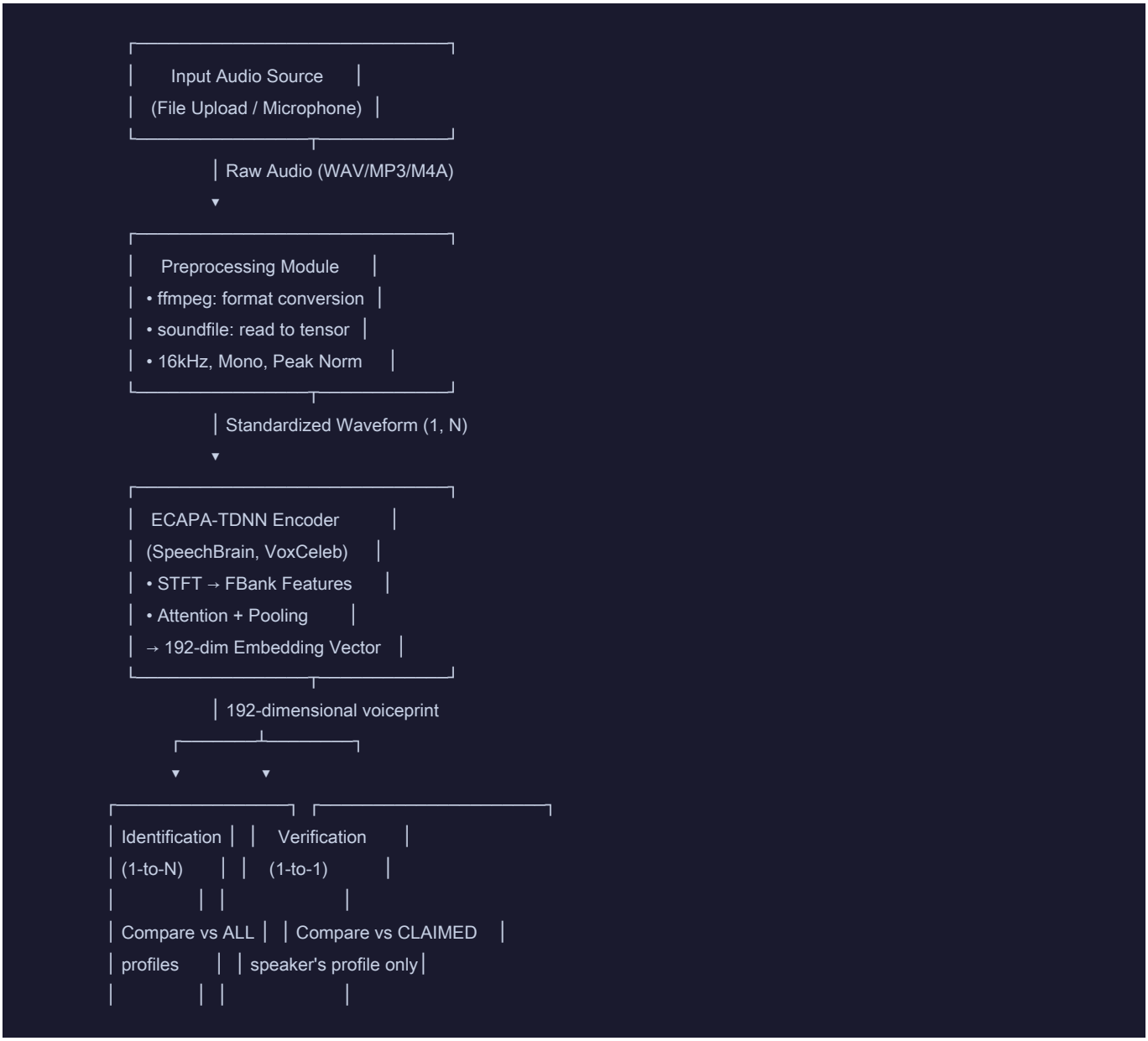
3. Chronological Evolution of the Field

Understanding the history helps frame our design choice:

Era	Approach	Key Idea	Limitation
1995–2005	GMM-UBM	Statistical voice modeling	Fragile, not compact
2005–2013	i-Vectors	Compact fixed-size representat	Handcrafted features, slow
2014–2017	d-Vectors (DNN)	Neural embeddings	Shallow, small data
2018–2019	X-Vectors (TDNN)	Deep neural embeddings + aug	Good baseline
2020–Present	**ECAPA-TDNN** (Ours)	Channel attention + aggregatio	**State of the art**

4. System Design

4.1 System Architecture (Block Diagram Description)





4.2 Software Components

Component	Technology	Why Chosen
Core Language	Python 3.14	Dominant language for ML; rich ecos
Neural Network	SpeechBrain ECAPA-TDNN	State-of-the-art, pretrained, easil
Tensor Operations	PyTorch	Industry standard deep learning fra
Audio I/O	`soundfile` (libsndfile)	Python 3.14-compatible, no deprecat
Format Conversion	`ffmpeg` (subprocess)	Universal, handles M4A/MP3/WAV/FLAC
Similarity Metric	NumPy cosine similarity	Fast, deterministic, well-understoo
Evaluation	`scikit-learn`	Standard ML evaluation library
Visualization	`matplotlib`	Industry-standard plotting
Web Interface	Streamlit	Python-native, browser-based UI, mi

4.3 Hardware Requirements

Minimum (Development & Demo):

- Any modern laptop/desktop with macOS, Linux, or Windows
- 4 GB RAM
- A microphone (built-in or USB)
- Internet connection (only needed once, to download the pretrained model ~85 MB)

For IoT Smart Lock Integration (Future Scope):

Approach	Hardware	Pros	Cons
Edge Computing	Raspberry Pi 4 + USB Mic + Re	No internet needed, Private	Higher cost (~Rs.6000)
Cloud API	ESP32 + I2S Mic (INMP441) +	Cheap (~Rs.1000), Simple	Needs Wi-Fi, Adds latency

4.4 Data Collection Protocol

- Speakers: 3 registered speakers (Advait, Omar, Parth)
- Sentences: 10 unique English sentences per speaker
- Repetitions: 3 recordings per sentence
- Total: 90 recordings (≈ 6 minutes of audio total)
- Formats: Raw files in M4A, MP3, and WAV; standardized to 16kHz mono WAV in preprocessing

Train/Test Split Strategy — Sentence-Disjoint:

- Reference set: Sentences S01–S07 (21 files per speaker = 63 total)
- Test set: Sentences S08–S10 (9 files per speaker = 27 total)

We deliberately used a sentence-disjoint split. This means the test recordings use different sentences than the reference recordings. This avoids the model "cheating" by recognizing the content of what was said, rather than the voice itself — making it a true text-independent speaker recognition test.

5. Mathematical Foundations

5.1 Short-Time Fourier Transform (STFT)

Before feeding audio into the neural network, the raw waveform (time-domain signal) is converted to a spectrogram using the STFT. A 25ms window is slid across the audio (with 10ms hop), and the FFT is applied to each window:

$$X(t, f) = \sum_{n=0}^{N-1} x(n) \cdot w(n-t) \cdot e^{-j2\pi fn/N}$$

This tells us which frequencies (f) are present at each moment in time (t). Mel-scale filterbanks then compress this into ≈80 FBank features per frame, mimicking human hearing.

5.2 Cosine Similarity

When comparing two embedding vectors A (query voice) and B (stored profile):

$$\text{Cosine Similarity} = \frac{A \cdot B}{||A|| \cdot ||B||}$$

- $A \cdot B$ = **Dot Product**: measures directional alignment
- $||A||$ = **L2 Norm** (Euclidean length of vector A)
- Result range: **[-1, 1]** where 1 = identical, 0 = unrelated, -1 = opposite

We use cosine similarity (not Euclidean distance) because it is magnitude-invariant — it is unaffected by how loud or soft the recording is, only by the direction of the voiceprint vector in 192-dimensional space.

5.3 Peak Normalization

All waveforms are normalized: waveform = waveform / max(|waveform|). This ensures the peak amplitude is exactly ±1.0, making scores fair across quiet and loud recordings.

6. Results Summary

Metric	Value
Total Test Recordings	27
Correctly Identified	27
Accuracy	**100%**
Precision (macro)	1.00
Recall (macro)	1.00
F1-Score (macro)	1.00
Avg. Genuine Similarity	0.8136
Avg. Impostor Similarity	0.4462
Gap (genuine – impostor)	0.37 (large, ideal)

The large gap between genuine and impostor similarity scores (0.81 vs 0.45) demonstrates that the ECAPA-TDNN embeddings are highly discriminative for our 3-speaker dataset.

7. Future Scope

7.1 Short-Term Improvements

- Add the 4th speaker to validate scalability
- Calibrate the UNKNOWN threshold statistically by collecting impostor recordings and plotting DET/ROC curves to find the Equal Error Rate (EER)
- Add more speakers to stress-test the system (50, 100, 1000 speakers)

7.2 Medium-Term Enhancements

- Anti-Spoofing / Liveness Detection: Integrate an ASVspoof-based model that detects whether audio is a live human or a replayed/AI-generated recording. This is a critical security layer for any biometric authentication system.
- Real-Time Streaming: Process audio in streaming 2-second chunks for instant feedback, rather than waiting for the full recording to finish.

7.3 Long-Term / Research Scope

- Speaker Diarization: "Who Spoke When?" — segmenting a multi-speaker conversation (meeting recording, podcast) and labeling segments per speaker.
- Cross-Lingual & Noise Robustness Analysis: Evaluate whether the system recognizes speakers when they switch languages (Hindi to English), whisper, or speak in a noisy café. Adapt using fine-tuning if needed.
- IoT Smart Lock: Integrate with ESP32 (cloud API approach) or Raspberry Pi (edge approach) and a relay module to build a voice-activated physical door lock system.

8. References

1. Reynolds, D. A., & Rose, R. C. (1995). Robust text-independent speaker identification using Gaussian mixture speaker models. *IEEE Trans. Speech Audio Process.*, 3(1), 72–83.

Link: IEEE Xplore

2. Dehak, N., Kenny, P. J., Dehak, R., Dumouchel, P., & Ouellet, P. (2011). Front-end factor analysis for speaker verification. *IEEE Trans. Audio Speech Lang. Process.*, 19(4), 788–798.

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Link: Google Research PDF

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3830–3834.** *(Primary model used in this project)*

Link: arXiv:2005.07143 | ISCA Archive

6. Ravanelli, M., Parcollet, T., Plantinga, P., et al. (2021). SpeechBrain: A General-Purpose Speech Toolkit. *arXiv:2106.04624*. *(Toolkit used in this project)*

Link: arXiv:2106.04624 | GitHub